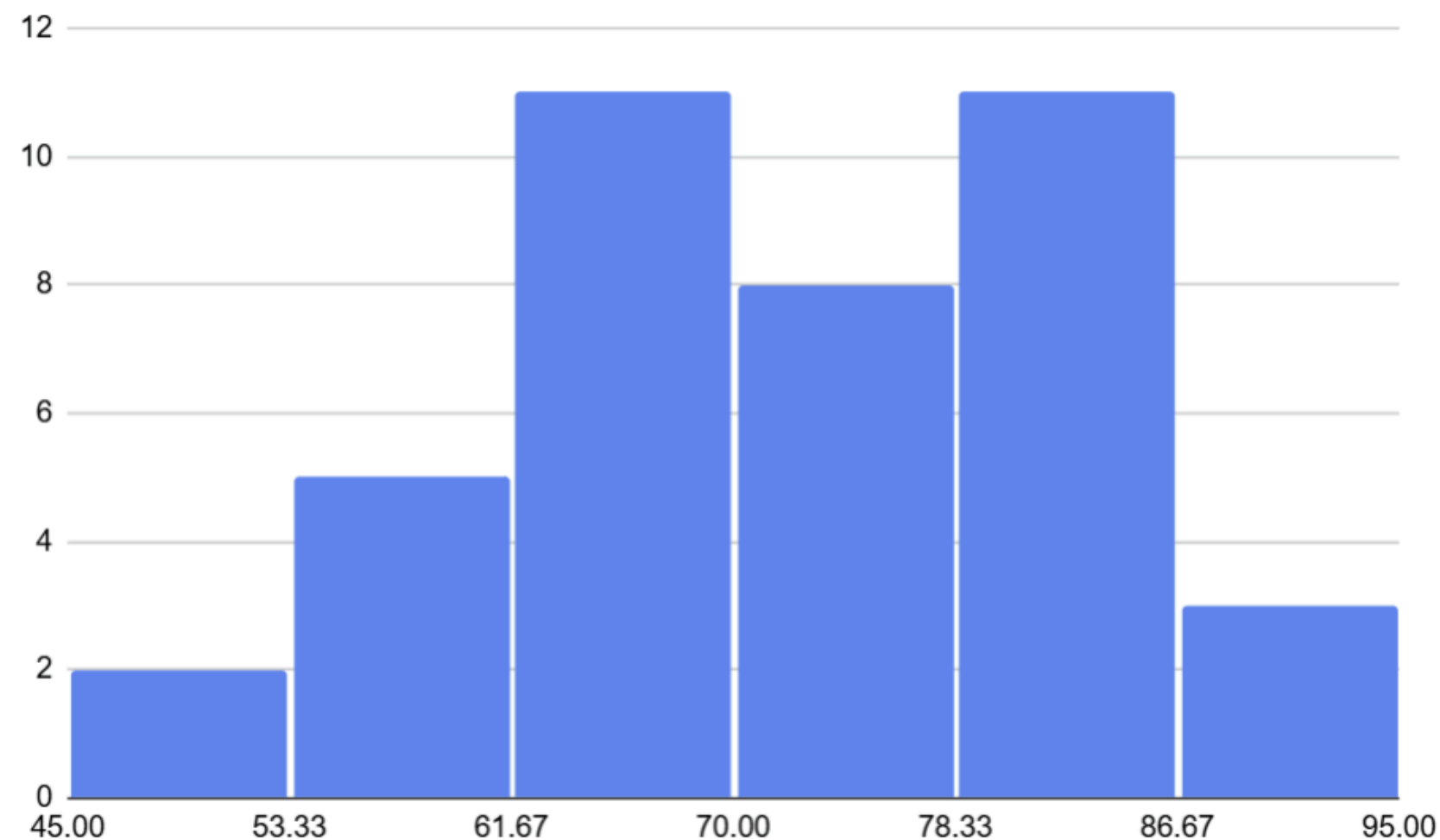


1. (a) H0: the scores are normally distributed; Ha: the scores are not normally distributed.

(b) sample mean (and other stats are calculated on Google Sheets) = 72.38, std = 11.08;

(c) I'm sorry that I have no idea how to do (c)

(d) based on the chart, it's close to a normal distribution



2. After generating 200 random numbers, separate the interval sections into A, B, C, D, and E:

A: 33; B: 37; C:37; D:48; E:45

apply chi-square test: $\frac{49}{40} + \frac{9}{40} + \frac{9}{40} + \frac{64}{40} + \frac{25}{40} = 3.9$, which is smaller than $\chi^2(0.05, 4) = 9.488$

The repeated tests have also not rejected this.

3. H0: Degree and industry are independent, Ha: They are dependent

$\chi^2 = \sum \frac{(O - E)^2}{E}$... Calculated with Google Sheets, we get $\chi^2 \approx 11.11$

$df = (2-1)(4-1) = 3$; CV = 11.345

$11.11 < 11.345 \rightarrow$ fail to reject H0

4. (a) independence, normality, and equal variance.
(b) ANOVA assumes the data is normally distributed
(c) ANOVA is useful for analyzing more than 3 factors, examples include medical analysts comparing drug effects, HR comparing workers' effectiveness, etc.

5. (a) H_0 : All means equal; H_a : At least one differs from the rest

	SS	df	MS	F
Between	8420	3	2807	4.52
Within	9920	16	620	
total	18340	19		

- (c) $F = 4.52$; $CV \approx 3.24 < F$, reject H_0
(d) comparing each data' s means $\rightarrow$ only C differs significantly from others.

6. (a) H_0 : All treatment have equal means; H_a : At least one differs

(b) $SST = SSTR + SSBL + SSE$

→ $SST = 130 + 264 + 220 = 614$

(c) $LSD \approx 5.9$

after comparing means, none are significant